Check for updates

Advanced Functional Materials

ADVANCED FUNCTIONAL MATERIALS
www.afm-journal.de

RESEARCH ARTICLE

Ballistic-Like Thermal Transport Between Fillers in Highly Conductive Stretchable Nanocomposites

C. Muhammed Ajmal[1] | Seongsu Cheon[2] | Yeongbin Kim[2,3] | Hongdeok Kim[2,3] | Joonmyung Choi[2,3] | Seunghyun Baik[1,2]

[1]Center for Nanotubes and Nanostructured Composites, Sungkyunkwan University, Suwon, Republic of Korea | [2]School of Mechanical Engineering, Sungkyunkwan University, Suwon, Republic of Korea | [3]SKKU National Lab for Intelligent Energy Solution Technology (SIEST), Sungkyunkwan University, Suwon, Republic of Korea

Correspondence: Seunghyun Baik (sbaik@skku.edu)

Received: 2 June 2026 | **Revised:** 10 August 2026 | **Accepted:** 15 August 2026

Keywords: ballistic-like thermal transport | barrier height | barrier width | nanocomposites | thermal conductivity

ABSTRACT

Heat transport across a potential energy barrier is largely unexplored in bulk nanocomposites. Here we report ballistic-like thermal transport between fillers in conductive nanocomposites, where thermal conductivity (κ) increases as channel length (i.e., barrier width, δ_B) is increased by stretching, maintaining strain-invariant thermal conductance. This is enabled by in situ generated silver nanosatellite (AgNS) particles (3.4 nm), realizing a short δ_B (4.1 nm). The barrier height (λ_B) is also tuned to be negligible by crosslinking silicone rubber matrix. The κ (21.94 $Wm^{-1}K^{-1}$) is also considerably greater than the uncrosslinked nanocomposite without AgNS particles. The stretching does not introduce additional scattering between AgNS particles, since δ_B is smaller than phonon mean free path (9.3 nm), although scattering exists in the bulk nanocomposite. The transport changes depending on δ_B and λ_B, resulting in diffusive transport ($\delta_B \geq 1.3$ μm, regardless of λ_B) and tunneling-dominated ($\delta_B < 5$ nm, non-negligible λ_B) regimes. The ballistic-like thermal transport can be utilized for flexible electronics and biomedical applications where strain-invariant heat dissipation is required.

1 | Introduction

Energy (like electron or phonon) transport across a potential barrier is of fundamental interest for electrically or thermally conductive devices and materials. Electron tunneling occurs through a thin insulating potential energy barrier due to its wave-like nature [1–3]. The tunneling probability depends exponentially on barrier width (δ_B) and height (λ_B), making the tunneling current extremely sensitive to small variations in these parameters [2, 3]. Classical approximations such as the Simmons' theory and Wentzel-Kramers-Brillouin method have successfully described the tunneling current-voltage relationship and barrier dependence [1, 2, 4]. This principle has been extensively explored in device physics forming the basis for tunnel diodes, resonant tunneling transistors, scanning tunneling microscopy, and magnetic tunnel junctions [5–9].

On the other hand, electron tunneling is relatively less explored in conductive nanocomposites although it governs charge transport when fillers are separated by nanometer gaps. Nanocomposites are typically composed of conductive nanoparticulate fillers dispersed in an insulating polymer, where long-range conduction pathways are constructed through the fillers above percolation threshold [3, 10–13]. Nevertheless, a significant fraction of fillers

C. Muhammed Ajmal and Seongsu Cheon contributed equally to this work.

All rights reserved, including rights for text and data mining and training of artificial intelligence technologies or similar technologies.
© 2026 Wiley-VCH GmbH

still remains isolated, and electron tunneling occurs when the interparticle distance is sufficiently small (e.g., < 10 nm) [2, 3, 13–15]. In such case, the electrical conductivity (σ) of nanocomposites is critically influenced by δ_B and λ_B [2, 3, 13]. The random nature of particle size and dispersion makes the precise control of δ_B and λ_B challenging, thereby limiting rigorous tunneling analysis in bulk-scale nanocomposites [2, 3, 13]. We recently reported silver nanosatellite (AgNS) particles by the in situ etching and reduction reaction of micro-scale silver flakes (AgFLs) in polymer matrix, enabling the precise control of particle size (< 5 nm) and δ_B (< 10 nm) [3, 13, 15]. The λ_B was also accurately modulated by cross-linking silicone rubber (SR) matrix [3]. This enabled the systematic electron tunneling study in bulk-scale nanocomposites, where the tunneling behavior was dependent on δ_B and λ_B [2, 3, 13]. Surprisingly, the strain-invariant electrical resistance was observed when δ_B was small (< 10 nm) and λ_B was negligible [3, 13]. The resistance was invariant even when δ_B was increased over 50% by stretching [3].

Heat transport across a potential energy barrier is largely unexplored in conductive nanocomposites. Tunneled electrons can also carry heat in electrically conductive nanocomposites, where the Lorenz number (L) quantitatively relates electronic thermal conductivity (κ_e) to σ via the Wiedemann-Franz law ($\kappa_e/\sigma T = L$, T is temperature) [16–18]. However, the dependence of phonon thermal conductivity (κ_p) on δ_B and λ_B remains an open question, where the total thermal conductivity (κ) is the sum of κ_e and κ_p. Quasi-ballistic thermal transport was observed at nanoscale interfaces, where the channel length was smaller than the mean free path [19]. An exceptionally large heat flux across nanometer vacuum gaps, beyond the existing theories, was also observed [20]. Low-frequency acoustic phonons were found to tunnel through the subnanometer vacuum gap by coupling to evanescent electric fields, exhibiting phonon tunneling behavior [21, 22].

Here we investigate the heat transport of bulk-scale nanocomposites in 3 distinct regimes, depending on δ_B and λ_B. The δ_B is precisely controlled by the in situ generated AgNS particles (diameter = 3.4 nm), and λ_B (0, 0.07, and 0.13 eV) is tuned by the gap state energy level of cross-linked SR matrix. In the diffusive regime ($\delta_B \geq 1.3$ µm, regardless of λ_B), κ decreases as δ_B is increased by stretching. The variation is primarily due to the effective volumetric filler fraction modulation by stretching, which could be precisely described by the percolation theory. In the tunneling regime ($\delta_B < 5$ nm, non-negligible λ_B), κ decreases rapidly as δ_B is increased. Electron tunneling dominates this regime, showing exponential decay in both σ and κ_e with regard to δ_B, which is accurately modeled by the Simmons approximation theory for quantum tunneling. Surprisingly, ballistic-like thermal transport between fillers is observed when δ_B is small (< 5 nm) and λ_B is negligible (0 eV). Both κ and σ increase when δ_B is increased by stretching, maintaining constant thermal (G_t) and electrical (G_e) conductances. Molecular dynamics (MD) simulation reveals that κ_p increases with stretching due to the enhanced Ag-SR interfacial vibrational coupling and polymer chain alignment. The δ_B is smaller than both phonon and electron mean free paths (9.3 and 11.4 nm), enabling the strain-invariant G_t and G_e. The κ is also significantly higher in this regime (21.9 W m^{-1} K^{-1}), compared with that (0.41 W m^{-1} K^{-1}) of the uncrosslinked nanocomposite in the diffusive transport regime, in spite of the identical Ag concentration (40 vol%). This work first demonstrates that ballistic-like thermal transport between fillers can be realized in the bulk nanocomposite by carefully controlling δ_B and λ_B. There is no additional scattering even when δ_B is increased by stretching, although scatterings exist in the bulk nanocomposite.

2 | Results and Discussion

2.1 | The Energy Barrier Width and Height Modulation in Thermally Conductive Nanocomposites

Electron tunneling occurs in a metal-insulator-metal junction when the insulator is sufficiently thin (Figure 1a). The δ_B is equivalent to the interparticle distance in conductive nanocomposites. The Fermi levels are aligned when the materials are in contact at equilibrium. For an ideal perfect insulator, λ_B is the difference between the work function of Ag (Φ_{Ag}) and electron affinity of insulator (χ_{SR}) [3, 23]. However, synthesized polymer (e.g., SR) has energy levels within the bandgap (τ_{SR}) due to defects, chemical impurities, and crosslinked structures [3, 24–26]. The gap states would form a conduction path if δ_B is sufficiently small (< 10 nm), resulting in $\lambda_B = \Phi_{Ag_}\tau_{SR}$ [3]. When the applied voltage across the junction (V) is sufficiently low, the tunneling current density (J) can be described by the Simmons approximation theory [2, 3, 13].

$$J = \frac{3(2m_e\lambda_B)^{\frac{1}{2}}}{2\delta_B}\left(\frac{e}{h}\right)^2 V \exp\left(-\frac{4\pi\delta_B}{h}(2m_e\lambda_B)^{\frac{1}{2}}\right) \quad (1)$$

where m_e is the effective mass of an electron, h is Planck's constant, and e is the elementary charge [2, 3]. The acronyms and symbols used in this study are summarized in Table S1.

Figure 1b shows a scanning electron microscopy (SEM) image of the nanocomposite synthesized by dispersing AgFLs in crosslinked SR (AgFL-SR$_{DDTP_7wt.\%}$, AgFL = 40 vol%). The crosslinking agent is 2,5-dimethyl-2,5-di(tert-butylperoxy)hexane (DDTP), and the relative concentration of DDTP in SR is denoted in the subscript. The average size of AgFLs is 4.3 µm (Figure S1), and the average distance between AgFLs (i.e., δ_B) is 1.3 µm. An atomic force microscopy (AFM) image shows the smooth surface of AgFLs (Figure 1b inset). The AgNS particles are generated by the in situ etching and reduction reaction of AgFLs using tetrahydrofuran (THF) peroxide (0.045 M), following a previously published protocol (see Methods for details) [3, 13, 15]. Figure 1c shows an SEM image of the AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposite (AgFL-AgNS = 40 vol%). Note that the total Ag concentration is invariant, even after the reaction, since AgNS particles are generated from AgFLs. The surface of AgFLs becomes rough after the reaction (Figure 1c inset), and medium AgNS particles are generated. A magnified high-resolution transmission electron microscopy (HRTEM) image shows uniformly distributed small AgNS particles, which play a critical role in tunneling (Figure 1d) [3, 13]. The average sizes of medium and small AgNS particles are 135 and 3.8 nm, respectively (Figure S2). The average distance between small AgNS particles (i.e., δ_B) is 4.1 nm. The δ_B is further modulated by stretching the nanocomposite. The specimen becomes anisotropic after

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

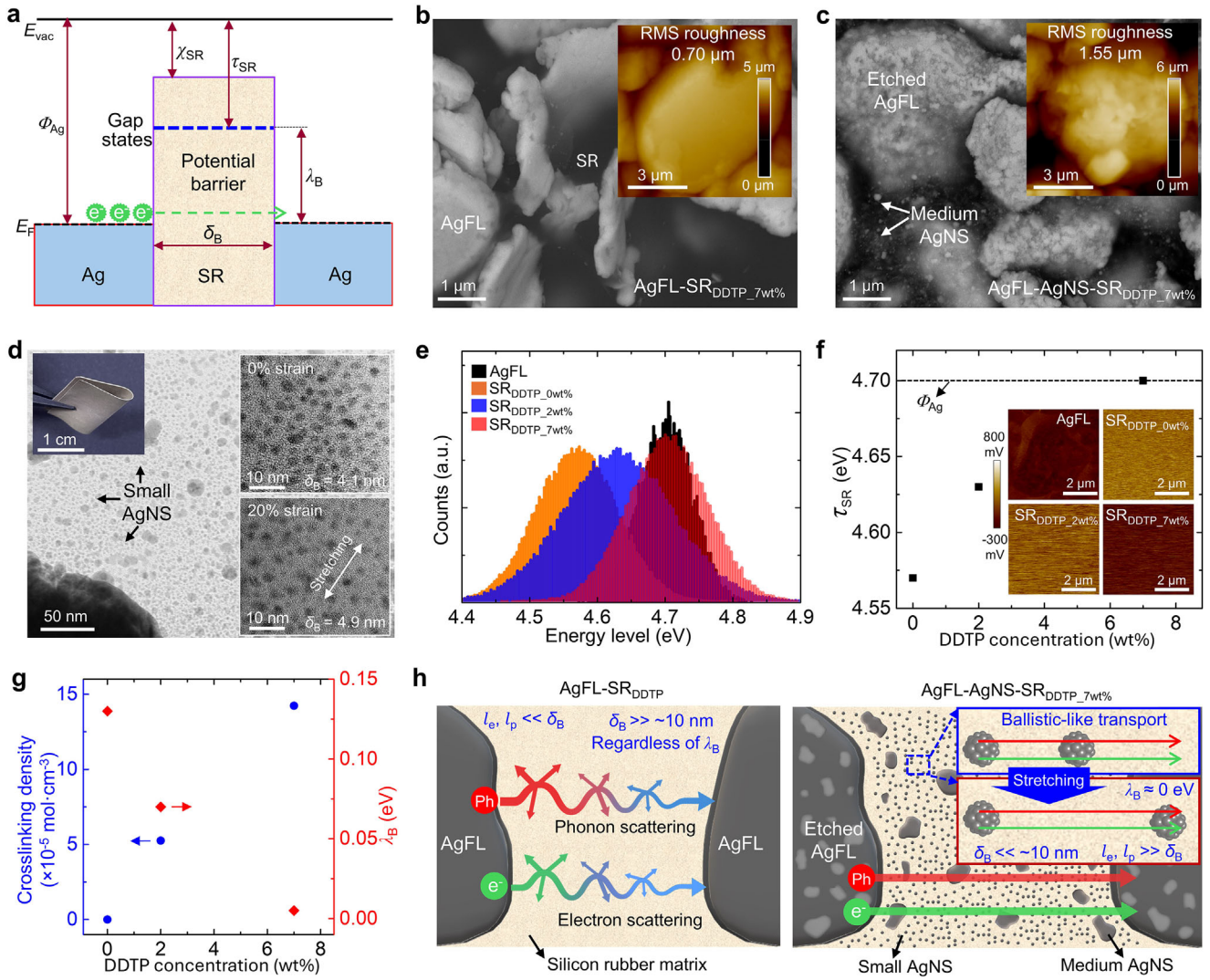


FIGURE 1 | The energy barrier width and height modulation in thermally conductive AgFL-AgNS-SR$_{DDTP}$ nanocomposites. (a) Ag-SR-Ag tunneling junction diagram. (b,c) Cross-sectional SEM images of the AgFL-SR$_{DDTP_7wt.\%}$ (THF peroxide = 0 M) and AgFL-AgNS-SR$_{DDTP_7wt.\%}$ (THF peroxide = 0.045 M) nanocomposites. The total Ag concentration is 40 vol% in both nanocomposites. AFM images of large AgFLs are provided as insets. (d) HRTEM images of the AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposite (AgFL-AgNS = 40 vol%). Magnified images at 0 and 20% strain are also shown. An optical image of the bent nanocomposite is provided in the inset. (e) KPFM analysis: Φ_{Ag} and τ_{SR} (SR$_{DDTP_0wt.\%}$, SR$_{DDTP_2wt.\%}$, and SR$_{DDTP_7wt.\%}$). (f) The mean τ_{SR} as a function of the DDTP concentration in SR (0, 2, and 7 wt.%). The corresponding contact potential difference mapping images are also provided. (g) The crosslinking density and λ_B as a function of the DDTP concentration in SR (0, 2, and 7 wt.%). (h) Schematics of electron and phonon transport pathways in the AgFL-SR$_{DDTP}$ (δ_B >> ~10 nm, regardless of λ_B) and AgFL-AgNS-SR$_{DDTP_7wt.\%}$ (δ_B << ~10 nm, λ_B = 0 eV) nanocomposites.

stretching, and δ_B is 4.9 nm along the stretched direction at 20% strain (Figure 1d inset). Note that δ_B varies with THF peroxide concentration, and it is optimized at 0.045 M in our previous electron tunneling investigation [3].

Figure 1e,f shows Φ_{Ag} and τ_{SR} measured by Kelvin probe force microscopy (KPFM). AgFLs and smooth pure SR films (DDTP = 0, 2, or 7 wt.%) are placed on a silicon wafer for the measurement with a chromium-platinum coated silicon tip. It is calibrated using highly ordered pyrolytic graphite (HOPG, work function = 4.6 eV) [3]. The contact potential difference (V_{CPD}) and corresponding topography mapping images are provided in Figure 1f inset and Figure S3. The average Φ_{Ag} is 4.70 eV. Interestingly, τ_{SR} increases from 4.57 to 4.70 eV as the DDTP concentration increases from 0 to 7 wt.% [3]. This is attributed to the formation of additional gap states caused by the crosslinking-induced new bond formation and change in crystallinity [3, 27, 28].

Figure 1g shows the crosslinking density, measured by the equilibrium swelling method, as a function of the relative DDTP concentration in SR [3, 29, 30]. The crosslinking density significantly increases as the DDTP concentration increases, reaching 1.42×10^{-4} mol•cm^{-3} at DDTP = 7 wt.%. This indicates that the strong C—C covalent bond is successfully formed by DDTP [3]. The elastic modulus also increases as the DDTP concentration increases (Figure S4) [3]. The AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposite shows a higher elastic modulus than the AgFL-SR$_{DDTP_7wt.\%}$ nanocomposite owing to more effective stress distribution in the presence of AgNS particles [3, 15]. The λ_B decreases from 0.13 to 0.07 and 0 eV as the DDTP concentration increases from 0 to 2

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and 7 wt.% (Figure 1g) [3]. A nearly perfect overlap between Φ_{Ag} and τ_{SR} leads to a negligible λ_B (0 eV) at DDTP = 7 wt.%.

Figure 1h briefly outlines the research strategy of this work. The electron and phonon transport in conductive nanocomposites can be classified using δ_B and λ_B, in conjunction with electron and phonon mean free paths (l_e and l_p). The δ_B is significantly greater than l_e and l_p when it is large enough (>> ~10 nm), as in the case of AgFL-SR_{DDTP} nanocomposite (Figure 1h left). The electron and phonon transport should be diffusive, regardless of λ_B [3, 15]. In contrast, when δ_B is sufficiently small (<< ~10 nm), it can be smaller than l_e and l_p as in the AgFL-AgNS-SR_{DDTP} nanocomposite. In such case, the transport becomes dependent on λ_B. Surprisingly, the ballistic-like thermal transport between fillers is observed when λ_B is negligible (0 eV) as shown in Figure 1h right. The σ and κ increase as δ_B increases, maintaining constant G_t and G_e, as will be discussed later.

2.2 | Thermal and Electrical Transport Analysis

Figure 2a shows the κ of the AgFL-SR_{DDTP} and AgFL-AgNS-SR_{DDTP} nanocomposites as a function of the relative DDTP concentration in SR. The κ is measured by the laser flash method (see Methods for details) [17, 31, 32]. The total Ag concentration is identical (40 vol%), and the nanocomposite becomes brittle when the filler concentration is increased further [3]. The λ_B decreases as the DDTP concentration increases as discussed in Figure 1g. The κ of the uncrosslinked AgFL-$SR_{DDTP_0wt.\%}$ nanocomposite is only 0.41 $Wm^{-1}K^{-1}$. This is partly due to voids in the uncrosslinked SR matrix [3, 13]. The κ significantly increases to 7.27 $Wm^{-1}K^{-1}$ with the addition of DDTP crosslinker (AgFL-$SR_{DDTP_2wt.\%}$) due to the strong C-C bond formation and removal of voids. A further increase in the DDTP concentration does not increase κ (AgFL-$SR_{DDTP_7wt.\%}$) when δ_B is large (1.3 μm). In contrast, the κ of the AgFL-AgNS-SR_{DDTP} nanocomposite continuously increases from 0.94 to 14.95 and 21.94 $Wm^{-1}K^{-1}$ as the DDTP concentration increases from 0 to 2 and 7 wt.%. The κ is more dependent on λ_B when δ_B is small (4.1 nm).

The carrier concentration of the nanocomposites is high (>10^{21} cm^{-3}) at Ag = 40 vol%, as will be discussed later. The σ of the nanocomposites is also characterized since the heat carried by electrons (κ_e) is significant for the total κ (Figure 2a inset). The σ is measured by the in-line four-point probe method (Figure S5) [11]. A similar changing trend of σ is observed, compared with κ, with regard to the DDTP concentration [3]. The in-plane and out-of-plane κ values of all the nanocomposites are similar, demonstrating isotropic transport property (Figure 2b). The detailed measurement method is described in Figure S6.

The δ_B is further modulated by stretching the nanocomposite. The stretching inevitably changes the effective volumetric filler concentration (Figure 2c). The total volume of the specimen changes during stretching, whereas the volume of solid Ag fillers remains invariant (Figure 2c inset) [11, 33]. A detailed calculation method is provided in Note S1. This, in turn, changes the density (ρ) and specific heat (C_p) of the nanocomposite as a function of strain. The experimentally measured ρ of the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite shows a close agreement with the theoretical value calculated by the rule of mixture, indicating a negligible void fraction (Figure 2c). The strain-dependent ρ variation is calculated using the effective Ag filler concentration and rule of mixture. It is then slightly adjusted to agree with the experimental value at 0% strain. The ρ decreases from 4.57 to 4.34 gcm^{-3} as the effect Ag concentration decreases from 40 to 38 vol% at 20% strain. The strain-dependent C_p variation is also calculated using the same method (Figure S7). The strain-dependent ρ and C_p are used to obtain κ (= $\rho\alpha C_p$), where α is diffusivity.

The α of the nanocomposites is measured by the laser flash method without graphite coating. The thickness and α values of all the specimens are provided in Table S2. Three specimens are measured for each condition, and the mean values and standard deviation are provided in Figure 2. Figure S8 shows the optical, SEM, and AFM images of the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite. The clamped specimen does not show any mechanical deformation (Figure S8a). Note that the specimen is also clamped using holders for the laser flash measurement. The cross-sectional SEM image shows a smooth surface and uniform thickness (Figure S8b). The root-mean-square roughness obtained from the AFM area scan is 0.4 μm (Figure S8c). The specimen is stretched stepwise, and the in-plane α is measured as a function of strain by the laser flash method using a ring-shaped sample holder (Figure S9). This yields an average of the α values in the stretched and perpendicular directions. The raw in-plane α data are also provided in Figure S10. The experimental data are well described by a fitting curve. The laser pulse width (0.3 ms) is significantly shorter than the half-rise time (249 ms), demonstrating the reliability of the measurement. The signal decay after reaching the maximum is due to the subsequent heat dissipation away from the measurement region [34]. The α values in the stretched and perpendicular directions need to be analyzed separately in the future. The uncertainty analysis is also carried out (Note S2). The measurement uncertainty of κ is found to be 3.01%.

The in-plane κ and σ of the nanocomposites are investigated at 3 distinctive regimes, depending on δ_B and λ_B: diffusive, tunneling, and ballistic-like transport regimes. Figure 2d shows the normalized κ and σ of the AgFL-$SR_{DDTP_7wt.\%}$ nanocomposite (δ_B = 1.3 μm at 0% strain, λ_B = 0 eV) as a function of strain. Both κ and σ decrease as δ_B is further increased by strain. The AgFL-$SR_{DDTP_2wt.\%}$ nanocomposite with a nonnegligible λ_B (δ_B = 1.3 μm at 0% strain, λ_B = 0.07 eV) also shows a similar decreasing trend (Figure S11). The in-plane current of the nanocomposite is continuously measured by the two-point probe method (V = 10 mV), and σ is calculated using the geometry information (Note S3). The σ is also measured along the stretched direction using the in-line four-point probe method while stretching the specimen stepwise (Figure S5). The σ values obtained by two different methods are in a close agreement, demonstrating reliability of the analysis (Figure S12). The thermal and electrical transport in this regime is diffusive, regardless of λ_B, since δ_B (> 1.3 μm) far exceeds l_e and l_p as will be discussed later. In the diffusive transport regime, the filler concentration is a crucial factor determining the κ and σ of nanocomposites. Figure 2e shows the κ and σ of the AgFL-$SR_{DDTP_7wt.\%}$ nanocomposite as a function of the Ag filler concentration. The logistic regression percolation model shows an excellent fit to the data in a wide range of filler concentrations (see Note S4 for model details) [33]. The logistic regression percolation model, obtained in Figure 2e, is also applied to the

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

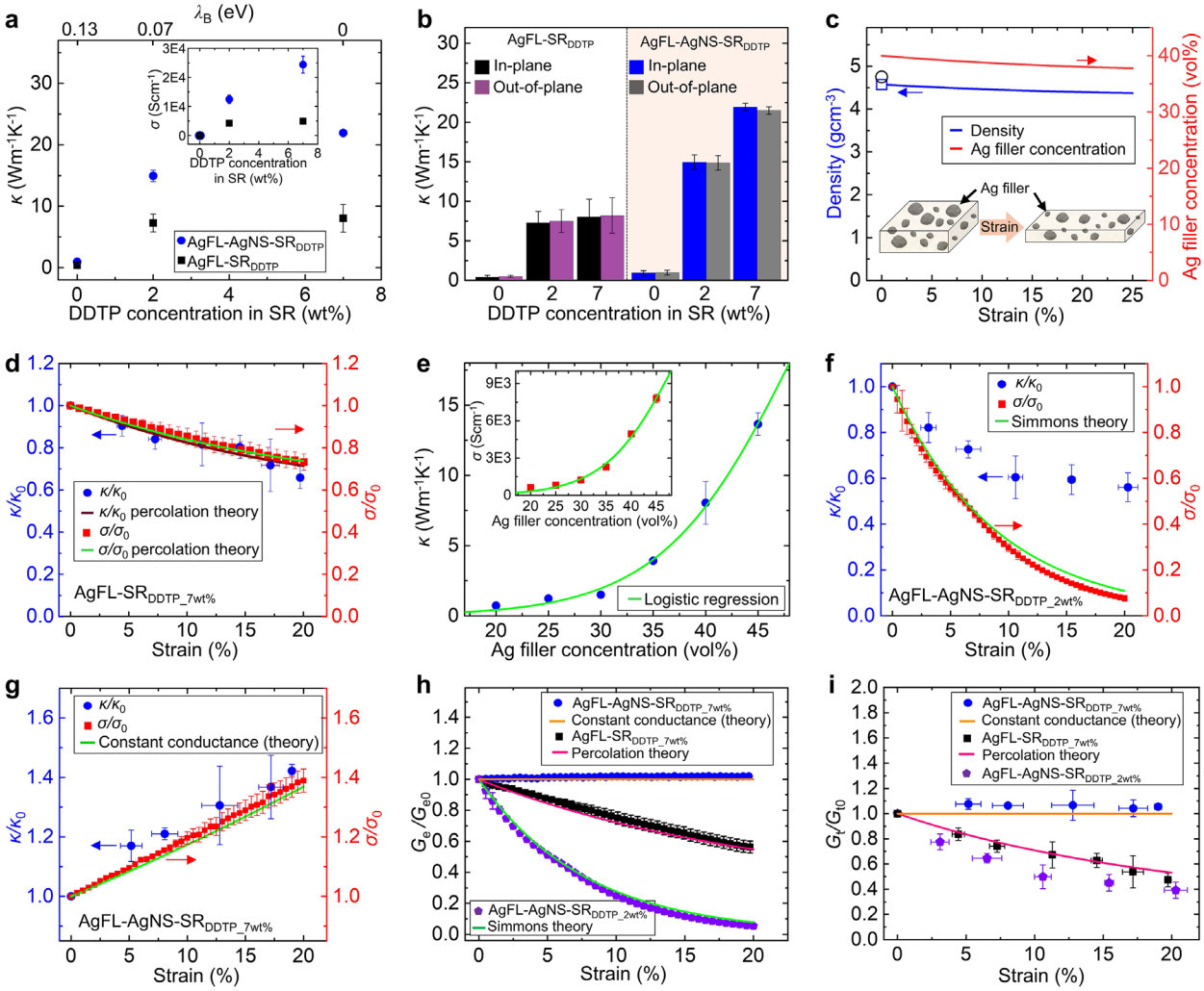


FIGURE 2 | Thermal and electrical transport analysis. (a) The κ and σ (inset) of the AgFL-SR$_{DDTP}$ and AgFL-AgNS-SR$_{DDTP}$ nanocomposites (Ag = 40 vol%) as a function of the DDTP concentration in SR (SR$_{DDTP_0wt.\%}$, SR$_{DDTP_2wt.\%}$, and SR$_{DDTP_7wt.\%}$). Three different specimens are investigated at each condition, and the error bar indicates the standard deviation of data. (b) The in-plane and out-of-plane κ of the AgFL-SR$_{DDTP}$ and AgFL-AgNS-SR$_{DDTP}$ nanocomposites (Ag = 40 vol%). (c) The effective Ag filler concentration and density of the AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposite are shown as a function of tensile strain. The density at 0% strain is experimentally measured (blue square), and the strain-dependent density variation is calculated using the effective Ag filler concentration. The nanocomposite density calculated by the rule of mixture is also shown (black circle). The inset illustrates the volume change of a rectangular specimen before and after stretching. (d) The normalized κ and σ of the AgFL-SR$_{DDTP_7wt.\%}$ nanocomposite as a function of strain. The σ is continuously measured during stretching (crosshead speed = 1 mm/min, applied voltage = 10 mV). The specimen is stretched stepwise for the κ measurement. The prediction of percolation theory (logistic regression model) is shown. (e) The κ and σ (inset) of the AgFL-SR$_{DDTP_7wt.\%}$ nanocomposite as a function of AgFL concentration. The logistic regression model fitting is also shown. (f) The normalized κ and σ of the AgFL-AgNS-SR$_{DDTP_2wt.\%}$ nanocomposite as a function of strain. The Simmons approximation theory prediction is shown. (g) The normalized κ and σ of the AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposite as a function of strain. The constant conductance line is shown. (h,i) The normalized electrical and thermal conductances of the AgFL-SR$_{DDTP_7wt.\%}$, AgFL-AgNS-SR$_{DDTP_2wt.\%}$, and AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposites as a function of strain. The predictions from constant conductance, percolation theory, and Simmons theory are shown.

data in Figure 2d because the stretching inevitably changes the effective filler concentration. The data in Figure 2d are well described by the percolation model. This implies that the major mechanism for the conductivity variation in diffusive regime is the stretching-induced effective filler concentration modulation. The scatterings also increase in the stretched channel between AgFLs (δ_B = 1.56 μm at strain = 20%).

Figure 2f shows the normalized κ and σ of the AgFL-AgNS-SR$_{DDTP_2wt.\%}$ nanocomposite (δ_B = 4.1 nm at 0% strain, λ_B = 0.07 eV). The δ_B is dramatically decreased by the AgNS particles. The transport falls in the tunneling-dominated regime when δ_B is sufficiently small with a non-negligible λ_B. The σ decreases exponentially as δ_B is increased by stretching, which is precisely described by the Simmons approximation theory for quantum tunneling Equation (1) [2, 3, 13]. It is interesting to note that the decreasing rate of κ, with increasing δ_B, is smaller than that of σ. This is due to the opposite variation trend of κ_e and κ_p, with regard to δ_B, as will be discussed later.

Figure 2g shows the normalized κ and σ of the AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposite (δ_B = 4.1 nm at 0% strain, λ_B = 0 eV).

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Surprisingly, both κ and σ increase as δ_B is increased by stretching, following the theoretical constant conductance line. Figure 2h,i summarizes the G_e and G_t of nanocomposites as a function of strain. The G_e is directly obtained from the current-strain relationship which is continuously measured during stretching by the two-probe method (Note S3). The in-plane κ is measured while stretching a rectangular specimen stepwise, and G_t is calculated using the κ and geometry information. Both G_e and G_t of the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite are invariant even when δ_B is increased by stretching. This phenomenon occurs only when δ_B is sufficiently small and λ_B is negligible. The δ_B (4.9 nm) is still less than l_e and l_p, even at 20% strain, as will be discussed shortly. Since the κ of the matrix is negligible, an effective κ of an Ag filler network is experimentally measured, which is mainly determined by the network structure and interfacial thermal conductance between neighboring fillers (i.e., Ag/SR/Ag). When the distance between neighboring fillers is less than l_p, the interfacial thermal conductance does not obey diffusive transport regime. This unique behavior is described as 'ballistic-like thermal transport between fillers' because it resembles the typical ballistic transport characteristics. The scattering occurs at the Ag/SR interfaces, where a mismatch in the phonon spectra can lead to a non-negligible interfacial thermal resistance. Nevertheless, the stretching does not change the conductance of the nanocomposite without introducing additional scattering, even though scattering occurs at the Ag/SR interfaces. In contrast, both G_e and G_t decrease with increasing strain in the diffusive regime (AgFL-$SR_{DDTP_7wt.\%}$). The transport is well-described by the logistic regression percolation model [33]. In the tunneling regime (AgFL-AgNS-$SR_{DDTP_2wt.\%}$), both G_e and G_t show fastest decay with regard to δ_B. The prediction by the Simmons approximation theory shows an excellent agreement with the experimentally measured G_e (see Note S5 for model details) [2, 3].

2.3 | Mean Free Path and Transport Mechanism Analysis

Figure 3a shows the carrier concentration (n) and mobility (μ) of the nanocomposites measured by the Van der Pauw method. The n exceeds 10^{21} cm^{-3} in all the nanocomposites (Ag = 40 vol%), demonstrating metal-like nature (Figure 3a) [32, 35, 36]. The μ of the AgFL-AgNS-$SR_{DDTP_2wt.\%}$ nanocomposite (3.83 $cm^2 V^{-1}s^{-1}$) in the tunneling regime is significantly smaller than that of the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite (17.88 $cm^2V^{-1}s^{-1}$) in the ballistic-like transport regime due to its higher λ_B (0.07 eV), in spite of the similar δ_B (4.1 nm), as shown in Figure 3a inset [37]. The kinetic theory, originally developed for a homogeneous metallic solid, is employed as an approximation to estimate the mean free paths of the nanocomposites because their σ and n are sufficiently high, demonstrating a metallic nature (e.g., $\sigma = 2.44 \times 10^4$ S cm^{-1} and $n = 8.52 \times 10^{21}$ cm^{-3} for AgFL-AgNS-$SR_{DDTP_7wt.\%}$) [32, 35, 36].

$$l_e = \Upsilon V_F,\ \sigma = \frac{e^2 n \Upsilon}{m_e},\ E_F = \frac{1}{2} m_e {V_F}^2 \tag{2}$$

where γ is the relaxation time, V_F is the Fermi velocity, e is the electron charge, m_e is the electron mass, and E_F is the Fermi energy level. The γ is calculated using the experimentally measured σ and n. The V_F is obtained from E_F, which is measured by ultraviolet photoelectron spectroscopy (Figure S13). As shown in Figure 3b, the l_e of the AgFL-$SR_{DDTP_7wt.\%}$ nanocomposite is 8.8 nm, which is significantly smaller than its δ_B (1.3 µm), confirming the diffusive transport regime. In the AgFL-AgNS-$SR_{DDTP_2wt.\%}$ nanocomposite ($\lambda_B = 0.07$ eV), the l_e (2.55 nm) is smaller than its δ_B (4.1 nm), and the δ_B-sensitive tunneling transport occurs. The l_e (11.4 nm) of the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite (λ_B = 0 eV) is considerably larger than its δ_B (4.1 nm). This analysis is also supported by an independent experiment where another AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite with a greater initial δ_B (8.3 nm) is synthesized using a lower THF peroxide concentration (0.038 M), while maintaining the negligible λ_B (Figure S14). A smaller number of AgNS particles is generated at a lower THF peroxide concentration, increasing δ_B [3, 13]. The G_e remains invariant up to 18% strain at which point δ_B becomes 9.8 nm. The G_e starts to decrease when the nanocomposite is stretched further, indicating that the specimen moves out of the ballistic-like transport regime at a greater δ_B (> 9.8 nm). This threshold is similar to the l_e (11.4 nm) obtained by the kinetic theory, demonstrating the reliability of the analysis. The mean free path needs to be analyzed more rigorously using a theoretical model for composites in the future.

The Wiedemann-Franz law is based on the free electron model of metals, which assumes that free electrons are the primary carriers of both heat and electrical current [16, 38]. In this study, the Wiedemann-Franz law is used to estimate the electronic contribution to thermal conductivity as an approximation, since the σ and n of the nanocomposites are sufficiently high (e.g., $\sigma = 2.44 \times 10^4$ S cm^{-1} and $n = 8.52 \times 10^{21}$ cm^{-3} for AgFL-AgNS-$SR_{DDTP_7wt.\%}$, as shown in Figure 3a). Furthermore, the Lorenz number (L) is experimentally obtained by precisely measuring the σ, μ, Seebeck coefficient (S), and reduced Fermi energy (η) as a function of temperature (Figure S15). The L is then obtained using the scattering parameter (r) and η (see Note S6 for details) [39, 40]. Interestingly, the experimentally measured L is identical to the theoretical Sommerfeld value (2.44×10^{-8} V^2 K^{-2}), confirming the metallic nature of the nanocomposites (Figure 3c inset) [38]. The κ_p is then obtained by κ—κ_e (= κ—$L\sigma T$) as an approximation. The κ_e is the dominant heat transport mechanism, rather than κ_p, as shown in Figure 3c. This is consistent with the metal-like nature of nanocomposites, considering the high Ag concentration (40 vol%).

Figure 3d shows the l_p of the nanocomposites obtained by the kinetic theory [18, 41].

$$\kappa_p = \frac{1}{3} \rho C_p v_s l_p,\ v_s = \left[\frac{1}{3}\left(\frac{1}{{v_l}^3} + \frac{2}{{v_t}^3}\right)\right]^{-\frac{1}{3}} \tag{3}$$

where v_s, v_l, and v_t are the average, longitudinal, and transverse acoustic velocities. The v_l and v_t are measured by the ultrasonic pulse-through method (Figure S16). The l_p (9.3 nm) of the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite ($\lambda_B = 0$ eV) is also considerably larger than its δ_B (4.1 nm). Interestingly, the l_p is similar for all the nanocomposites (8-9.3 nm). It is not dependent on λ_B (= Φ_{Ag}-τ_{SR}), which is originally devised to interpret electron tunneling. A stricter concept of barrier height for phonon transport needs to be developed in the future.

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

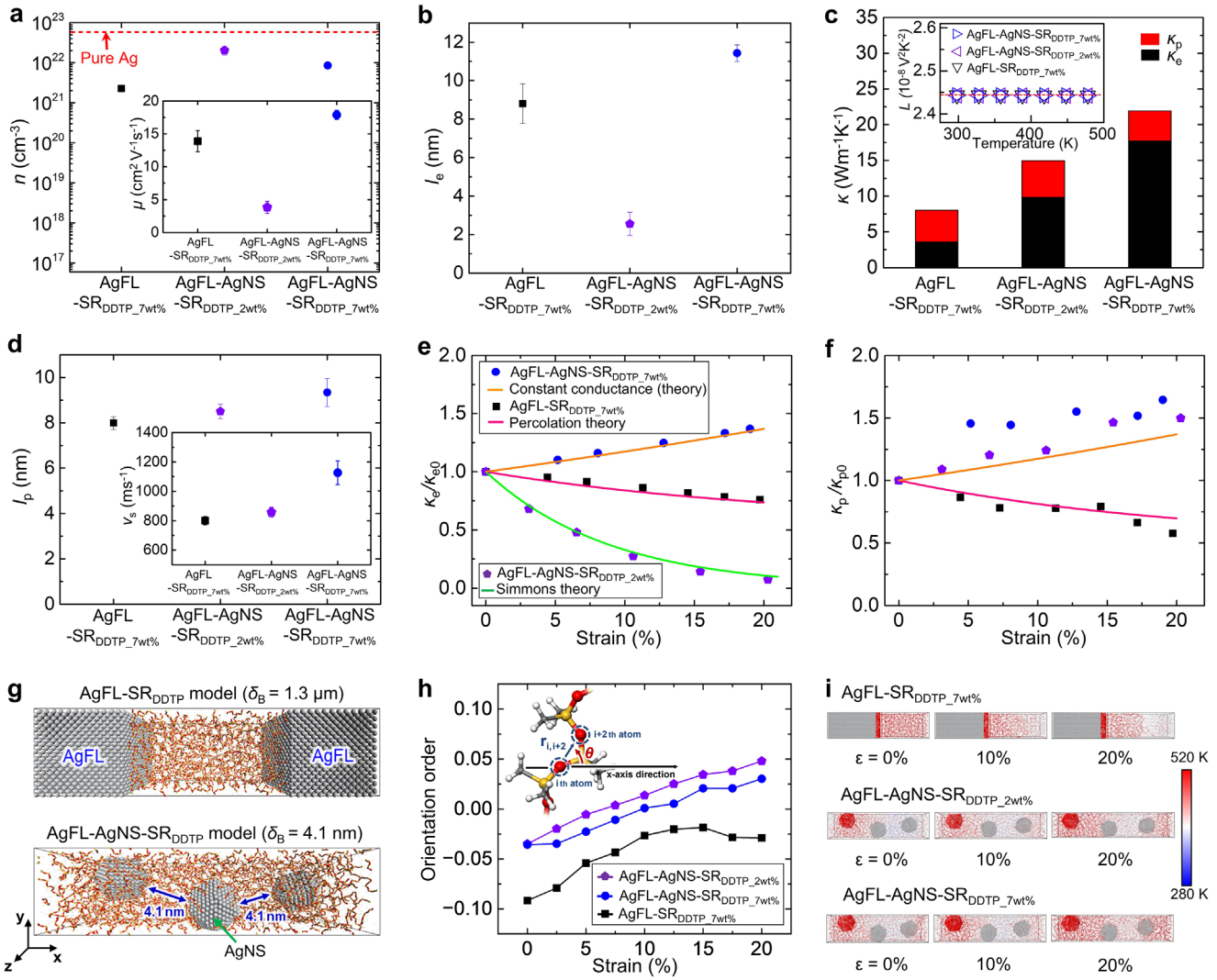


FIGURE 3 | Mean free path and transport mechanism analysis. (a,b) The carrier concentration, mobility, and electron mean free path of the AgFL-SR$_{DDTP_7wt.\%}$, AgFL-AgNS-SR$_{DDTP_2wt.\%}$, and AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposites. (c) The electronic thermal conductivity, phonon thermal conductivity, and Lorenz number (inset) of the AgFL-SR$_{DDTP_7wt.\%}$, AgFL-AgNS-SR$_{DDTP_2wt.\%}$, and AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposites. The dotted line indicates the Sommerfield value. (d) The phonon mean free path and sound velocity (inset) of the AgFL-SR$_{DDTP_7wt.\%}$, AgFL-AgNS-SR$_{DDTP_2wt.\%}$, and AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposites. (e,f) The normalized electronic and phonon thermal conductivities of the AgFL-SR$_{DDTP_7wt.\%}$, AgFL-AgNS-SR$_{DDTP_2wt.\%}$, and AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposites as a function of strain. The predictions from constant conductance, percolation theory, and Simmons theory are shown. (g) Representative atomistic models of the AgFL-SR$_{DDTP}$ and AgFL-AgNS-SR$_{DDTP}$ nanocomposites considered in MD simulations. (h) Evolution of the orientational order parameter of SR chains under uniaxial tension for the AgFL-SR$_{DDTP_7wt.\%}$, AgFL-AgNS-SR$_{DDTP_2wt.\%}$, and AgFL-AgNS-SR$_{DDTP_7wt.\%}$ models. (i) Representative temperature distributions at 500 ps obtained from thermal conduction simulations (tensile strain = 0%, 10%, and 20%).

Figure 3e,f shows the normalized κ_e and κ_p of the nanocomposites as a function of strain. The absolute values (κ, κ_e, and κ_p) are provided in Figure S17. The κ_e follows the same trend of σ, since it is derived from $\kappa_e = L\sigma T$ [16–18]. The theoretical predictions for κ_e are obtained by multiplying the theoretical predictions for σ by LT. The κ_p of the AgFL-SR$_{DDTP_7wt.\%}$ nanocomposite decreases as δ_B is increased by stretching. The decreasing trend could be precisely predicted by the percolation theory for κ_p (see Figure S18 for details). The decrease in the effective filler concentration is a major mechanism for the stretching-induced κ_p variation as discussed earlier. The phonon scattering between AgFLs should also increase by stretching since δ_B (1.3 μm) is significantly greater than l_p in the diffusive phonon transport regime. Surprisingly, the κ_p values of both AgFL-AgNS-SR$_{DDTP_2wt.\%}$ and AgFL-AgNS-SR$_{DDTP_7wt.\%}$ nanocomposites increase with stretching, following the theoretical constant conductance line. The ballistic-like phonon transport between fillers is observed, regardless of λ_B, indicating the absence of additional phonon scattering in the stretched channel. The δ_B (4.9 nm) of both nanocomposites is smaller than l_p even after stretching at 20%.

To elucidate the microscopic origin of the extraordinary phonon transport, three representative nanocomposite models (AgFL-SR$_{DDTP_7wt.\%}$, AgFL-AgNS-SR$_{DDTP_2wt.\%}$, and AgFL-AgNS-SR$_{DDTP_7wt.\%}$) are constructed within all-atom MD framework. As shown in Figure 3g, the AgFL-SR$_{DDTP}$ model is represented

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

by a bilayer structure consisting of an AgFL and an SR layer, whereas the AgFL-AgNS-SR_{DDTP} model is constructed with an interparticle distance of 4.1 nm reflecting the δ_B between AgNS particles. The simulations are designed to interpret differences in phonon transport through the SR chains by focusing on the Ag-SR interfacial interaction and chain deformation under uniaxial strain. The computational details are provided in Note S7.

The spatial distribution of Ag fillers significantly differs in the AgFL-SR_{DDTP} and AgFL-AgNS-SR_{DDTP} models. This results in distinct differences in polymer deformation under uniaxial tension and corresponding heat transfer characteristics. The Ag-SR interfacial interaction plays a critical role in restricting chain mobility and stabilizing the filler-matrix interface. As shown in Figure S19, the Ag-SR interfacial interaction energy density is higher in the AgNS containing models, indicating stronger adsorption of SR chains onto AgNS surfaces. Such enhanced adsorption is expected to suppress excessive chain slip during tensile loading and to promote efficient vibrational coupling across the interface [42]. Furthermore, the interparticle spacing (δ_B = 4.1 nm) is sufficiently small compared to the characteristic dimension of SR chains, such that a single polymer chain can adsorb onto multiple AgNS particles [43]. This multi-point adsorption can mechanically bridge neighboring AgNS particles, inducing chain tensioning under deformation.

The structural difference between the AgFL-SR_{DDTP} and AgFL-AgNS-SR_{DDTP} models also has a significant impact on the microstructural response under deformation. As shown in Figure 3h, the SR chains in AgFL-$SR_{DDTP_7wt.\%}$ initially exhibit weak perpendicular alignment to the tensile direction. Although the SR chain orientational order increases with strain, a further alignment is limited beyond a certain strain level (10%–15%) due to the pronounced chain slip. Resultantly, the SR chains remain disordered even at 20% strain. In contrast, both the AgFL-AgNS-$SR_{DDTP_2wt.\%}$ and AgFL-AgNS-$SR_{DDTP_7wt.\%}$ models show a nearly linear increase in orientational order with increasing strain, indicating gradual and stable alignment along the loading direction.

The microstructural difference also has a direct correlation with heat conduction. The representative temperature distribution at 500 ps is shown in Figure 3i, and the full-time evolution (0 to 500 ps) is provided in Figure S20. When applying high temperature to the AgFL and monitoring temperature propagation through the SR matrix, the AgFL-$SR_{DDTP_7wt.\%}$ model exhibits a gradual decrease in temperature propagation with increasing strain. This is consistent with the diffusive phonon transport behavior. On the other hand, both the AgFL-AgNS-$SR_{DDTP_2wt.\%}$ and AgFL-AgNS-$SR_{DDTP_7wt.\%}$ models exhibit improved temperature propagation with increasing strain. The unique characteristics result from the stronger Ag-SR interfacial vibrational coupling and deformation-induced chain alignment [44].

2.4 | Device Application Demonstration of Ballistic-Like Thermal Transport Characteristics

The ballistic-like thermal transport characteristics (i.e., strain-invariant thermal conductance) of the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite is employed for the heat dissipation demonstration of a foldable smartphone. The uniform heat distribution between the left and right modules of a foldable smartphone is a critical design requirement. However, the conventional heat spreader has a severe thermal bottleneck at the hinge region, since it experiences tensile strain (10%–20%) during folding. The strain interrupts thermal conduction pathways, leading to localized heat accumulation and nonuniform temperature increase.

The nanocomposite is installed in a foldable smartphone mockup for application demonstration (Figure 4a,b). One end is attached to a ceramic heater using silver paste, emulating an application processor, and the other end is attached to a metal block (heat sink). As shown in Figure 4c, four thermocouples are employed to precisely monitor heat propagation: one on the heater (TC_1) and three on the nanocomposite (TC_2-TC_4). Figure 4d compares the temperatures of AgFL-$SR_{DDTP_7wt.\%}$ and AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposites as a function of heating time (unfolded flat position). The TC_2 of the AgFL-$SR_{DDTP_7wt.\%}$ nanocomposite increases as the heater is turned on. TC_3 is lower than TC_2 since heat propagates along the nanocomposite. Interestingly, the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite shows a more uniform temperature distribution than the AgFL-$SR_{DDTP_7wt.\%}$ nanocomposite (i.e., lower TC_2 and higher TC_3), even at flat position, due to the higher κ (Figure 2a).

The mockup phone is folded to 0, 90, 120, and 150° (Figure S21). The nanocomposite is heated for 30 min at each folding angle (heating power = 0.38 W), and steady state temperature data are provided in Figure 4e. The corresponding strain of the nanocomposite is 0, 7.7, 11.3, and 14.7%, respectively. Note that the folding angle 180° is excluded for analysis. Heat flows directly through the modules, which are in contact at 180°, rather than through the nanocomposite. The small contact resistance between the heater and nanocomposite results in similar TC_1 and TC_2 values. The TC_3 and TC_4 of AgFL-$SR_{DDTP_7wt.\%}$ nanocomposite decrease as the folding angle increases. The AgFL-$SR_{DDTP_7wt.\%}$ nanocomposite ($\delta_B \geq 1.3$ μm) is in the diffusive transport regime. The scattering increases and G_t decreases as the specimen is stretched by folding (Figure 2i), decreasing TC_3 and TC_4. In contrast, the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite shows a more uniform temperature (i.e., smaller difference between TC_1 and TC_4), compared with the AgFL-$SR_{DDTP_7wt.\%}$ nanocomposite. Furthermore, TC_3 and TC_4 are invariant, even when the specimen is stretched by folding, confirming the ballistic-like thermal transport of AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite ($\delta_B \leq$ ~4.8 nm, λ_B = 0 eV). The ballistic-like thermal transport characteristics can be employed for diverse application fields, including flexible electronics and biomedical devices, where strain-invariant heat dissipation is required under mechanical deformation.

3 | Conclusions

This work first reports the ballistic-like thermal transport between fillers in bulk nanocomposites. Three distinctive transport regimes are found by the investigation of heat transport across a potential energy barrier in bulk nanocomposites. In the diffusive regime ($\delta_B \geq 1.3$ μm, regardless of λ_B), κ decreases as δ_B is increased by stretching in accordance with the percolation

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

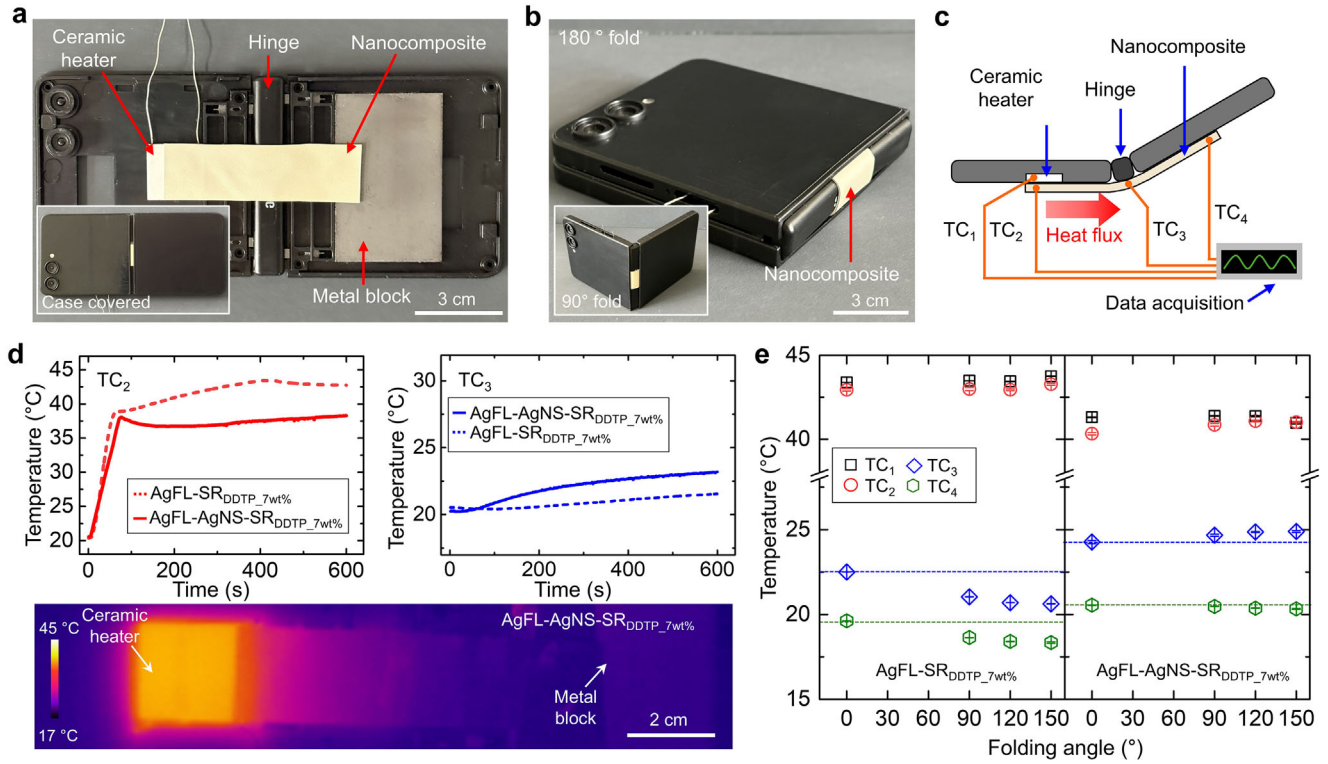


FIGURE 4 | Device application demonstration of ballistic-like thermal transport characteristics. (a) The AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite is installed in a foldable smartphone mockup. One end is attached to a ceramic heater using silver paste, and the other end is attached to a metal block (heat sink). The image of the mockup with covered case is provided in the inset. (b) Folded mockup images. (c) Thermocouples (TC_1-TC_4) are installed to precisely monitor heat propagation. (d) The temperature variation of the AgFL-$SR_{DDTP_7wt.\%}$ and AgFL-AgNS-$SR_{DDTP_7wt.\%}$ as a function of time (flat position). The heating power is identical (0.38 W). An infrared camera image of the AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposite is also shown (300 s after heating). (e) Steady-state temperature profiles of the AgFL-$SR_{DDTP_7wt.\%}$ and AgFL-AgNS-$SR_{DDTP_7wt.\%}$ nanocomposites as a function of folding angle.

theory. On the other hand, κ decreases rapidly with stretching in the tunneling regime ($\delta_B < 5$ nm, non-negligible λ_B). Electron tunneling dominates this regime, which could be precisely described by the Simmons approximation theory for quantum tunneling. Remarkably, ballistic-like thermal transport between fillers is observed when δ_B is small (< 5 nm) and λ_B is negligible (0 eV). The interfacial thermal conductance does not obey diffusive transport regime when δ_B is less than l_p. The κ increases when δ_B is increased by stretching, maintaining constant G_t. The stretching does not change the conductance of the nanocomposite without introducing additional scattering, even though scattering occurs at the Ag/SR interfaces. The MD simulation reveals that AgNS particles enhance interfacial vibrational coupling and SR chain alignment with stretching, improving phonon heat transport. The strain-invariant G_t of stretchable nanocomposites can provide new opportunities for the advanced thermal management of foldable/flexible electronics and biomedical devices.

4 | Methods

4.1 | Synthesis of the AgFL-AgNS-SR_{DDTP} and AgFL-SR_{DDTP} Nanocomposites

The AgFL-AgNS-SR_{DDTP} and AgFL-SR_{DDTP} nanocomposites are synthesized following a previously published protocol, and the procedure is briefly summarized as follows [3]. THF without a butylated hydroxytoluene (BHT) peroxidation inhibitor (Sigma-Aldrich, 401757) is air-bubbled for 72 h to generate THF peroxide (0.045 M) [3, 13]. SR (KCC Silicon Corporation, SH0010U, 15 wt.%) is dissolved in THF containing a BHT inhibitor (Sigma-Aldrich, 186562, THF peroxide = 0 M), and DDTP (KCC Silicon Corporation, 8 wt.%) is dissolved in hexane (Sigma-Aldrich, 178918) [3]. AgFLs (Metalor, SA-31812, 2.1 g) are then dispersed in the mixture of SR (2 g) and DDTP solutions with additional THF peroxide (15 mL) using tip sonication (420 W, 10 min). The relative concentration of DDTP in SR is 0, 2, or 7 wt.%. The mixture is further stirred (300 rpm, 60 min), drop-casted, and dried overnight at room temperature. Finally, the AgFL-AgNS-SR_{DDTP} nanocomposite is obtained by hot-pressing (170°C,3 MPa, 10 min) and curing (200°C, 4 h) [3]. THF with a BHT inhibitor is used, instead of THF peroxide, to prepare the AgFL-SR_{DDTP} nanocomposite [3].

4.2 | Characterization

The microstructure is analyzed by HRTEM (JEOL, Cs-corrected JEM-ARM 200F) and SEM (JEOL, JSM-7600F). The direction-dependent α is measured by the laser flash method using commercial in-plane and out-of-plane holders (NETZSCH, LFA467 HyperFlash) [45]. The ρ is measured by the Archimedes method (SARTORIOUS, Quintix224-1SKR), and C_p is measured by differential scanning calorimetry (NETZSCH, DSC214 Polyma, heating

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

rate: 5°C min^{-1}). The κ is then obtained by $\kappa = \rho\alpha C_p$ (specimen dimension = 15 × 15 × 0.18 mm^3). The σ is measured by the in-line four-point probe method with a geometry correction factor [11, 46, 47]. A current source (Keithley 6221) and a nanovoltmeter (Keithley 2182 A) are employed, and the distance between tungsten probes is 1 mm. The in-plane current of the nanocomposite is also recorded as a function of strain using a direct current power supply (Agilent, E3648A) and a data acquisition software (Keysight BenchVue) [3, 13]. The elastic modulus and Poisson's ratio are measured using a universal testing machine (Instron, 3343). The surface potential is analyzed by KPFM (Park Systems, NX10) using a chromium-platinum coated silicon tip. The μ, n, S are obtained by Hall effect (Lakeshore, HMS 8404) and Seebeck coefficient (NETZSCH, SBA458) measurement systems. The Fermi energy level is measured by ultraviolet photoelectron spectroscopy (Thermo Fisher Scientific, ESCALAB250Xi, monochromatic He-I radiation at 21.22 eV) [35]. The acoustic velocity is measured by the ultrasonic pulse-through method (Riteclab, UMS-100, 5 MHz). The application demonstration is carried out using a foldable smartphone mockup.

Author Contributions

C.M.A., S.C., and S.B. conceived and designed the experiments, which were carried out by C.M.A. and S.C. Y.K., H.K., and J.C carried out molecular dynamics simulation. C.M.A., S.C., and S.B. wrote the paper. All authors contributed to data analysis and scientific discussion.

Acknowledgements

This work was supported by the Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (No. 2021R1A6A1A03039696), the NRF grants funded by the Korea government (MSIT and MOE) (Nos. RS-2026-25501632 and RS-2025-23323210), the Korea Institute of Science and Technology (KIST) Institutional Program (No. 26Z9052), and the Start up Pioneering in Research and Innovation (SPRINT) through the Commercialization Promotion Agency for R&D Outcomes (COMPA) grant funded by the Korea government (MSIT) (No. RS-2025-02314328).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

1. R. Guo, L. Tao, M. Li, Z. Liu, et al., "Interface-Engineered Electron and Hole Tunneling," *Science Advances* 7 (2021): abf1033.

2. J. G. Simmons, "Generalized Formula for the Electric Tunnel Effect Between Similar Electrodes Separated by a Thin Insulating Film," *Journal of Applied Physics* 34 (1963): 1793–1803, https://doi.org/10.1063/1.1702682.

3. C. Muhammed Ajmal, J. Jeong, S. Cheon, M. K. Majee, H. Yang, and S. Baik, "Absence of Additional Stretching-Induced Electron Scattering in Highly Conductive Cross-Linked Nanocomposites With Negligible Tunneling Barrier Height and Width," *Advanced Science* 11 (2024): 2409337, https://doi.org/10.1002/advs.202409337.

4. A. Gruverman, D. Wu, H. Lu, et al., "Tunneling Electroresistance Effect in Ferroelectric Tunnel Junctions at the Nanoscale," *Nano Letters* 9 (2009): 3539–3543, https://doi.org/10.1021/nl901754t.

5. G. Binnig and H. Rohrer, "Scanning Tunneling Microscopy—From Birth to Adolescence," *Reviews of Modern Physics* 59 (1987): 615–625, https://doi.org/10.1103/RevModPhys.59.615.

6. S. Chakravarty and S. Kivelson, "Photoinduced Macroscopic Quantum Tunneling in Superconducting Interference Devices," *Physical Review Letters* 50 (1983): 1811–1814, https://doi.org/10.1103/PhysRevLett.50.1811.

7. A. A. Talin, K. A. Dean, and J. E. Jaskie, "Field Emission Displays: A Critical Review," *Solid-State Electronics* 45 (2001): 963–976.

8. A. Chanthbouala, A. Crassous, V. Garcia, et al., "Solid-State Memories Based on Ferroelectric Tunnel Junctions," *Nature Nanotechnology* 7 (2012): 101–104, https://doi.org/10.1038/nnano.2011.213.

9. V. Garcia and M. Bibes, "Ferroelectric Tunnel Junctions for Information Storage and Processing," *Nature Communications* 5 (2014): 4289, https://doi.org/10.1038/ncomms5289.

10. D. Jung, Y. Kim, H. Lee, et al., "Metal-Like Stretchable Nanocomposite Using Locally-Bundled Nanowires for Skin-Mountable Devices," *Advanced Materials* 35 (2023): 2303458.

11. K.-Y. Chun, Y. Oh, J. Rho, et al., "Highly Conductive, Printable and Stretchable Composite Films of Carbon Nanotubes and Silver," *Nature Nanotechnology* 5 (2010): 853–857, https://doi.org/10.1038/nnano.2010.232.

12. C. S. Boland, U. Khan, G. Ryan, et al., "Sensitive Electromechanical Sensors Using Viscoelastic Graphene-Polymer Nanocomposites," *Science* 354 (2016): 1257–1260, https://doi.org/10.1126/science.aag2879.

13. C. Muhammed Ajmal, S. Cha, W. Kim, K. P. Faseela, H. Yang, and S. Baik, "Invariable Resistance of Conductive Nanocomposite over 30% Strain," *Science Advances* 8 (2022): abn3365, https://doi.org/10.1126/sciadv.abn3365.

14. G. Lubineau, A. Mora, F. Han, I. N. Odeh, and R. Yaldiz, "A Morphological Investigation of Conductive Networks in Polymers Loaded with Carbon Nanotubes," *Computational Materials Science* 130 (2017): 21–38, https://doi.org/10.1016/j.commatsci.2016.12.041.

15. D. Suh, K. P. Faseela, W. Kim, et al., "Electron Tunneling of Hierarchically Structured Silver Nanosatellite Particles for Highly Conductive Healable Nanocomposites," *Nature Communications* 11 (2020): 2252, https://doi.org/10.1038/s41467-020-15709-8.

16. G. J. Snyder and E. S. Toberer, "Complex Thermoelectric Materials," *Nature Materials* 7 (2008): 105–114, https://doi.org/10.1038/nmat2090.

17. D. Suh, C. M. Moon, D. Kim, and S. Baik, "Ultrahigh Thermal Conductivity of Interface Materials by Silver-Functionalized Carbon Nanotube Phonon Conduits," *Advanced Materials* 28 (2016): 7220–7227, https://doi.org/10.1002/adma.201600642.

18. T. M. Tritt, *Thermal Conductivity: Theory, Properties, and Applications* (Springer, 2004), https://doi.org/10.1007/b136496.

19. M. E. Siemens, Q. Li, R. Yang, et al., "Quasi-Ballistic Thermal Transport From Nanoscale Interfaces Observed Using Ultrafast Coherent Soft X-Ray Beams," *Nature Materials* 9 (2010): 26–30.

20. K. Kloppstech, N. Könne, S.-A. Biehs, et al., "Giant Heat Transfer in the Crossover Regime Between Conduction and Radiation," *Nature Communications* 8 (2017): 14475, https://doi.org/10.1038/ncomms14475.

21. Y. Guo, C. Adessi, M. Cobian, and S. Merabia, "Atomistic Simulation of Phonon Heat Transport across Metallic Vacuum Nanogaps," *Physical Review B* 106 (2022): 085403, https://doi.org/10.1103/PhysRevB.106.085403.

22. V. Chiloyan, J. Garg, K. Esfarjani, and G. Chen, "Transition From Near-Field Thermal Radiation to Phonon Heat Conduction at Sub-Nanometre Gaps," *Nature Communications* 6 (2015): 6755, https://doi.org/10.1038/ncomms7755.

23. K. Mistry, M. Yavuz, and K. P. Musselman, "Simulated Electron Affinity Tuning in Metal-Insulator-Metal (MIM) Diodes," *Journal of Applied Physics* 121 (2017): 184504, https://doi.org/10.1063/1.4983256.

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

24. J. G. Simmons, "Theory of Metallic Contacts on High Resistivity Solids (II) Deep Traps," *Journal of Physics and Chemistry of Solids* 32 (1971): 2581–2591, https://doi.org/10.1016/S0022-3697(71)80104-X.

25. Y. Wang, S. Nasreen, D. Kamal, et al., "Tuning Surface States of Metal/Polymer Contacts toward Highly Insulating Polymer-Based Dielectrics," *ACS Applied Materials & Interfaces* 13 (2021): 46142–46150, https://doi.org/10.1021/acsami.1c12854.

26. T. Mizutani, Y. Takai, T. Osawa, and M. Ieda, "Barrier Heights and Surface States of Metal-Polymer (PET) Contacts," *Journal of Physics D: Applied Physics* 9 (1976): 2253–2259, https://doi.org/10.1088/0022-3727/9/15/016.

27. Z. Yan, K. Yang, Y. Zhang, S. Wang, and J. Li, "Crosslinking Dependence of Trap Distribution and Breakdown Performance of Crosslinked Polyethylene," *Journal of Materials Science: Materials in Electronics* 30 (2019): 20605–20613.

28. K. Yang, Y. Liu, Z. Yan, et al., "Enhanced Morphology-Dependent Tensile Property and Breakdown Strength of Impact Polypropylene Copolymer for Cable Insulation," *Materials* 13 (2020): 3935, https://doi.org/10.3390/ma13183935.

29. J. Schweitzer, S. Merad, G. Schrodj, F. Bally-Le Gall, and L. Vonna, "Determination of the Crosslinking Density of a Silicone Elastomer," *Journal of Chemical Education* 96 (2019): 1472–1478, https://doi.org/10.1021/acs.jchemed.8b00911.

30. P. J. Flory and J. Rehner Jr., "Statistical Mechanics of Cross-Linked Polymer Networks II. Swelling," *The Journal of Chemical Physics* 11 (1943): 521–526, https://doi.org/10.1063/1.1723792.

31. Z. Barani, A. Mohammadzadeh, A. Geremew, et al., "Thermal Properties of the Binary-Filler Hybrid Composites With Graphene and Copper Nanoparticles," *Advanced Functional Materials* 30 (2020): 1904008, https://doi.org/10.1002/adfm.201904008.

32. C. M. Ajmal, A. P. Benny, W. Jeon, S. Kim, S. W. Kim, and S. Baik, "In-situ Reduced Non-Oxidized Copper Nanoparticles in Nanocomposites With Extraordinary High Electrical and Thermal Conductivity," *Materials Today* 48 (2021): 59–71, https://doi.org/10.1016/j.mattod.2021.04.012.

33. S. Bae and S. Baik, "The Linear Response of Composite Pressure Sensors by Controlling Effective Volumetric Filler Concentration," *Sensors and Actuators A: Physical* 372 (2024): 115343, https://doi.org/10.1016/j.sna.2024.115343.

34. J. A. Cape and G. W. Lehman, "Temperature and Finite Pulse-Time Effects in the Flash Method for Measuring Thermal Diffusivity," *Journal of Applied Physics* 34 (1963): 1909–1913, https://doi.org/10.1063/1.1729711.

35. D. Suh, S. Lee, C. Xu, A. A. Jan, and S. Baik, "Significantly Enhanced Phonon Mean Free Path and Thermal Conductivity by Percolation of Silver Nanoflowers," *Physical Chemistry Chemical Physics* 21 (2019): 2453–2462, https://doi.org/10.1039/C8CP07229A.

36. Q. Zhu, S. Jiang, K. Ye, et al., "Hydrogen-Doping-Induced Metal-Like Ultrahigh Free-Carrier Concentration in Metal-Oxide Material for Giant and Tunable Plasmon Resonance," *Advanced Materials* 32 (2020): 2004059, https://doi.org/10.1002/adma.202004059.

37. S. Mukherjee, G. Samanta, M. N. Hasan, et al., "Achieving Nearly Barrier Free Transport in High Mobility ReS_2 Phototransistors with van der Waals Contacts," *npj 2D Materials and Applications* 8 (2024): 71.

38. N. W. Ashcroft and N. D. Mermin, *Solid State Physics* (Holt, Rinehart and Winston, 1976), 12.

39. L.-D. Zhao, J. He, C.-I. Wu, et al., "Thermoelectrics With Earth Abundant Elements: High Performance p-Type PbS Nanostructured With SrS and CaS," *Journal of the American Chemical Society* 134 (2012): 7902–7912, https://doi.org/10.1021/ja301772w.

40. D. Suh, S. Lee, H. Mun, et al., "Enhanced Thermoelectric Performance of $Bi_{0.5}Sb_{1.5}Te_3$-Expanded Graphene Composites by Simultaneous Modulation of Electronic and Thermal Carrier Transport," *Nano Energy* 13 (2015): 67–76, https://doi.org/10.1016/j.nanoen.2015.02.001.

41. Y. Xiao, C. Chang, Y. Pei, et al., "Origin of Low Thermal Conductivity in SnSe," *Physical Review B* 94 (2016): 125203, https://doi.org/10.1103/PhysRevB.94.125203.

42. J. Huang, J. Zhou, and M. Liu, "Interphase in Polymer Nanocomposites," *JACS Au* 2 (2022): 280–291.

43. D. J. Growney, O. O. Mykhaylyk, T. Derouineau, et al., "Star Diblock Copolymer Concentration Dictates the Degree of Dispersion of Carbon Black Particles in Nonpolar Media: Bridging Flocculation versus Steric Stabilization," *Macromolecules* 48 (2015): 3691–3704, https://doi.org/10.1021/acs.macromol.5b00517.

44. J. Liu and R. Yang, "Tuning the Thermal Conductivity of Polymers With Mechanical Strains," *Physical Review B* 81 (2010): 174122, https://doi.org/10.1103/PhysRevB.81.174122.

45. S. A. Abdul Jaleel, T. Kim, and S. Baik, "Covalently Functionalized Leakage-Free Healable Phase-Change Interface Materials With Extraordinary High-Thermal Conductivity and Low-Thermal Resistance," *Advanced Materials* 35 (2023): 2300956, https://doi.org/10.1002/adma.202300956.

46. C. M. Ajmal, S. Bae, and S. Baik, "A Superior Method for Constructing Electrical Percolation Network of Nanocomposite Fibers: In Situ Thermally Reduced Silver Nanoparticles," *Small* 15 (2019): 1803255.

47. F. M. Smits, "Measurement of Sheet Resistivities With the Four-Point Probe," *The Bell System Technical Journal* 37 (1958): 711–718.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: adfm77937-sup-0001-SuppMat.pdf.

16163028, 2026, 78, Downloaded from https://advanced.onlinelibrary.wiley.com/doi/10.1002/adfm.77937 by Joonmyung Choi - Pukyong National University , Wiley Online Library on [11/10/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License